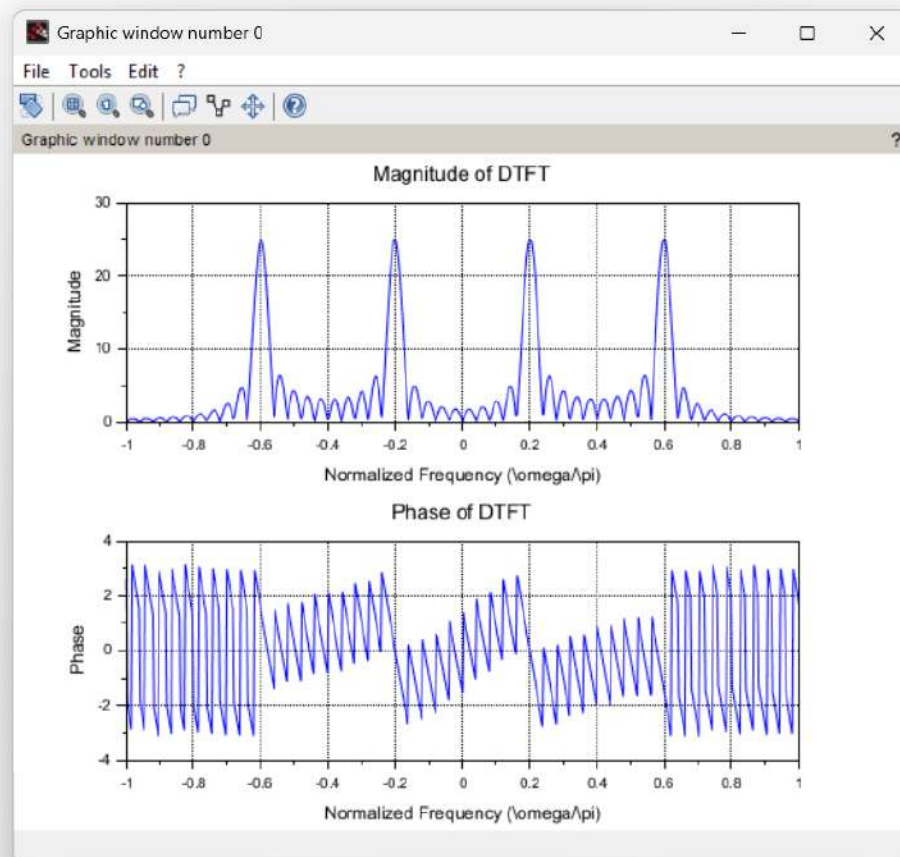


```

1 clc;
2 clear;
3 clf;
4
5 // Parameters
6 N = 50;
7 f0 = 0.1;
8 f1 = 0.3;
9
10 n = 0:N-1;
11
12 // Signal
13 x = cos(2*pi*f0*n) + sin(2*pi*f1*n);
14
15 // Frequency vector
16 omega = -pi:0.01:pi;
17
18 X = zeros(omega);
19
20 for k = 1:length(omega)
21     X(k) = sum(x.*exp(-1i.*omega(k).*n));
22 end
23
24 // Magnitude
25 subplot(2,1,1);
26 plot(omega/pi, abs(X));
27 xlabel("Normalized Frequency (\omega/\pi)");
28 ylabel("Magnitude");
29 title("Magnitude of DTFT");
30 grid();
31
32 // Phase
33 subplot(2,1,2);
34 plot(omega/pi, atan(imag(X), real(X)));
35 xlabel("Normalized Frequency (\omega/\pi)");
36 ylabel("Phase");
37 title("Phase of DTFT");
38 grid();
39

```



exp 8.3.sci (C:\Users\imjan\OneDrive\Documents\exp 8.3.sci) - SciNotes

File Edit Format Options Window Execute ?



exp 8.3.sci (C:\Users\imjan\OneDrive\Documents\exp 8.3.sci) - SciNotes

1.1 exp.sci X 1.2 exp.sci X exp 2.1.sci X 2.2.sci X exp 3.2.sci X 3.1.sci X 4.1 exp.sci X exp 4.2.sci X exp 5.1.sci X exp 5.2.sci X exp 6.1.sci X exp 6.2.sci X exp 7.1.sci X exp 7.3.sci X exp 8.3.sci X

```
1 // Define parameters
2 N = 31; // Total number of samples
3 n = 0:N-1; // Sample index
4 n1 = -10; // Start of rectangular pulse
5 n2 = 20; // End of rectangular pulse
6
7 // Define the rectangular pulse x[n]
8 x = zeros(1, N); // Initialize the signal with zeros
9 x(n1+1:n2+1) = 1; // Set values between n1 and n2 to 1
10
11 // Compute DTFT
12 omega = linspace(-%pi, %pi, 1000); // Frequency range
13 X = x * exp(-%i * n * omega); // DTFT calculation
14
15 // Plot the magnitude of DTFT
16
17 figure();
18 plot(omega/%pi, abs(X));
19 xlabel('Normalized Frequency - (\omega/\pi)');
20 ylabel('|X(e^{j\omega})|');
21 title('Magnitude of DTFT of Rectangular Pulse');
22
23 // Plot the phase of DTFT
24 figure();
25 plot(omega/%pi, atan(imag(X)./real(X)));
26 xlabel('Normalized Frequency - (\omega/\pi)');
27 ylabel('Phase of X(e^{j\omega})');
28 title('Phase of DTFT of Rectangular Pulse');
29
30 // Analyze the results in the Scilab console or report
31
```

